



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

AeroPulse : Aviation Data Analytics & Operational Insights

Domain : Data Analytics & Artificial Intelligence

Mentor :

Ms. Keerthana V. (Senior Executive - Business Analyst Exl Service)

Submitted By: Ms. Hanne Jenifer R

COURSE : DATA ANALYTICS

Project Timeline :

August 2026 to September 2026



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Declaration

The project titled “**AeroPulse — Aviation Data Analytics & Operational Insights**” has been mentored by **Ms. Keerthana V**, organised by SURE Trust, from August 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Done By :

Hanne Jenifer R

Mentor:

Ms. Keerthana V.

Senior Executive

Business Analyst Exl Service

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

- AeroPulse — Aviation Data Analytics & Operational Insights is an aviation analytics project focused on understanding flight delays and identifying the operational factors that contribute to them. The project transforms raw aviation data into meaningful insights that can support better understanding of airline and airport performance.
- The project follows an end-to-end data analytics workflow, beginning with data preparation and cleaning, followed by exploratory data analysis to identify patterns, trends, and relationships within flight operations. Key aspects such as delays, airlines, airports, time-based patterns, and operational performance were analyzed to understand where and when delays are most significant.
- The cleaned and analyzed data was then used to develop an interactive Power BI dashboard that presents key performance indicators, delay analysis, comparisons, trends, and operational insights in an accessible format. The project also includes a dedicated website that brings the AeroPulse analysis together into a more engaging and structured user experience, allowing the findings and insights to be communicated beyond the dashboard.
- Overall, AeroPulse demonstrates how aviation data can be transformed from raw operational records into actionable business intelligence. The project combines data analytics, visualization, and digital presentation to provide a clearer view of aviation delay patterns and operational performance, while demonstrating an end-to-end approach to solving a real-world analytics problem.



Introduction

Background and Context

The aviation industry generates large volumes of operational data through flight schedules, airline operations, airports, and delay records. Flight delays can affect passengers, airlines, and airport operations, making delay analysis an important area for understanding operational performance. AeroPulse uses aviation data analytics to examine these patterns and convert raw operational data into meaningful insights.

Problem Statement and Goals

Flight-delay data can be difficult to interpret when presented as raw records. The goal of AeroPulse is to analyze this data, identify significant delay patterns and operational trends, and present the findings in a clear and interactive manner. The project aims to help users understand when, where, and across which operational factors delays are more prominent.

Scope and Limitations

The scope of AeroPulse covers data preparation, exploratory data analysis, visualization, and interactive business intelligence using Power BI, supported by a dedicated project website. The analysis focuses on the information available in the selected dataset and therefore reflects only the period, variables, airlines, airports, and operational factors represented in that data. The project provides analytical insights and does not directly predict future delays or control airline operations.

Innovation Component

The innovation of AeroPulse lies in combining aviation data analytics with interactive business intelligence and a dedicated digital presentation layer. Instead of presenting analysis as static results, the project brings together data-driven insights, interactive Power BI visualizations, and a purpose-built website to make aviation operational insights easier to explore and understand.



Project Objectives

Objectives and Goals

The primary objectives of AeroPulse are to:

- Analyse aviation data to understand flight delay patterns and operational performance.
- Clean and prepare the dataset for reliable analysis.
- Perform exploratory data analysis to identify trends, patterns, and key factors associated with delays.
- Develop meaningful Key Performance Indicators (KPIs) to evaluate aviation operational performance.
- Build an interactive Power BI dashboard for exploring delay and performance insights.
- Present the project findings through a dedicated AeroPulse website in a clear and engaging manner.

Expected Outcomes and Deliverables

The project is expected to deliver:

- A cleaned and analysis-ready aviation dataset.
- Exploratory data analysis with relevant findings and visualizations.
- An interactive Power BI dashboard containing aviation performance and delay insights.
- A dedicated AeroPulse website for presenting the project, analysis, and findings.
- A consolidated set of data-driven insights that improves understanding of aviation delays and operational performance.



Methodology and Results

Methods and Technology Used

AeroPulse follows an end-to-end data analytics methodology. The project begins with data preparation and cleaning to improve data quality and consistency. Exploratory Data Analysis (EDA) is then performed to identify trends, patterns, and relationships within the aviation data. The processed data is subsequently used for visualization and interactive business intelligence.

The project methodology consists of the following stages:

Data Collection → Data Cleaning → Exploratory Data Analysis → Data Modelling & KPI Development → Power BI Visualization → Insights → Website Presentation

Tools and Software Used

The major tools and technologies used in the project are:

- **Python** – Data cleaning, preprocessing, and exploratory data analysis.
- **Pandas** – Data manipulation and transformation.
- **Matplotlib / Seaborn** – Data visualization during exploratory analysis.
- **Power BI** – Interactive dashboard development, KPI analysis, and business intelligence visualization.
- **GitHub** – Version control and project documentation.

Data Collection Approach

The project uses an aviation dataset containing flight and operational information required for delay analysis. The collected data was examined for data quality issues, missing values, inconsistencies, and irrelevant information before being prepared for analysis.



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The dataset was then transformed into an analysis-ready format through data cleaning and preprocessing. Relevant features were retained and additional analytical fields were derived where required to support the project's objectives.

Project Architecture

AeroPulse follows a structured data-to-insight architecture:

Raw Aviation Data



Data Cleaning & Preprocessing



Exploratory Data Analysis



Prepared Analytical Dataset



Power BI Data Model & KPIs



Interactive Dashboard



Insights & Findings



AeroPulse Website

This architecture separates data preparation, analysis, visualization, and presentation into distinct stages. It enables the raw aviation data to be progressively transformed into interactive analytical insights and a user-facing project experience.

Final Project Working Screenshots

The final working screenshots of the AeroPulse project are presented below. Each screenshot is accompanied by a brief explanation describing its purpose, key features, and the insights it provides.



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Aviation Performance & Flight Operations Analytics

Total Flights
5.41M
vs Last Year ▲ 15.4%

On-Time %
83.69%
vs Last Year ▼ 7.2%

Cancellation %
1.74%
vs Last Year ▲ 71.1%

Avg Delay (min)
10.72
vs Last Year ▲ 465.8%

Volume Growth %
15.44%
vs Last Year —

Select an area to explore the analytics



Flight Delay Analysis



Cancellations & Hotspots



Flight Operations & Efficiency

YEAR
☐ 2020
☒ 2021
☐ 2022
☐ 2023
☐ 2024
☐ 2025



Flight Delay Performance Analysis



Total Delayed Flights
1.07M
vs Last Year ▲ 103.1%

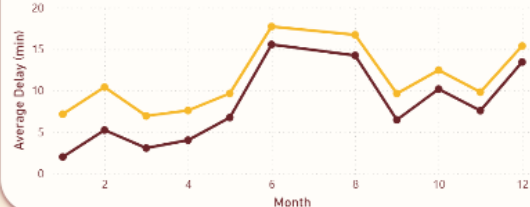
Avg Departure Delay (min)
8.58
vs Last Year ▲ 316.5%

Avg Arrival Delay (min)
11.59
vs Last Year ▲ 69.4%

Flights Delayed <15min
844.33K
0.00M 1.69M

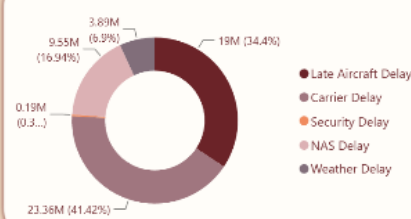
Average Arrival Delay Trend

● Avg Departure Delay (min) ● Avg Arrival Delay (min)

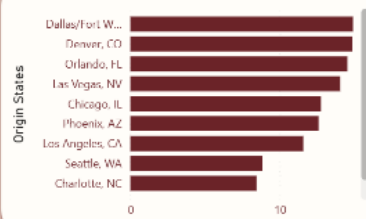


YEAR
☐ 2020
☒ 2021
☐ 2022
☐ 2023
☐ 2024
☐ 2025

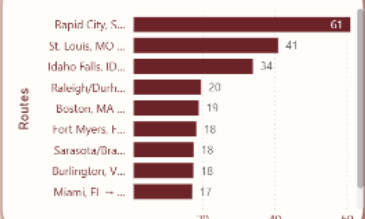
Flights Affected by Delay Cause

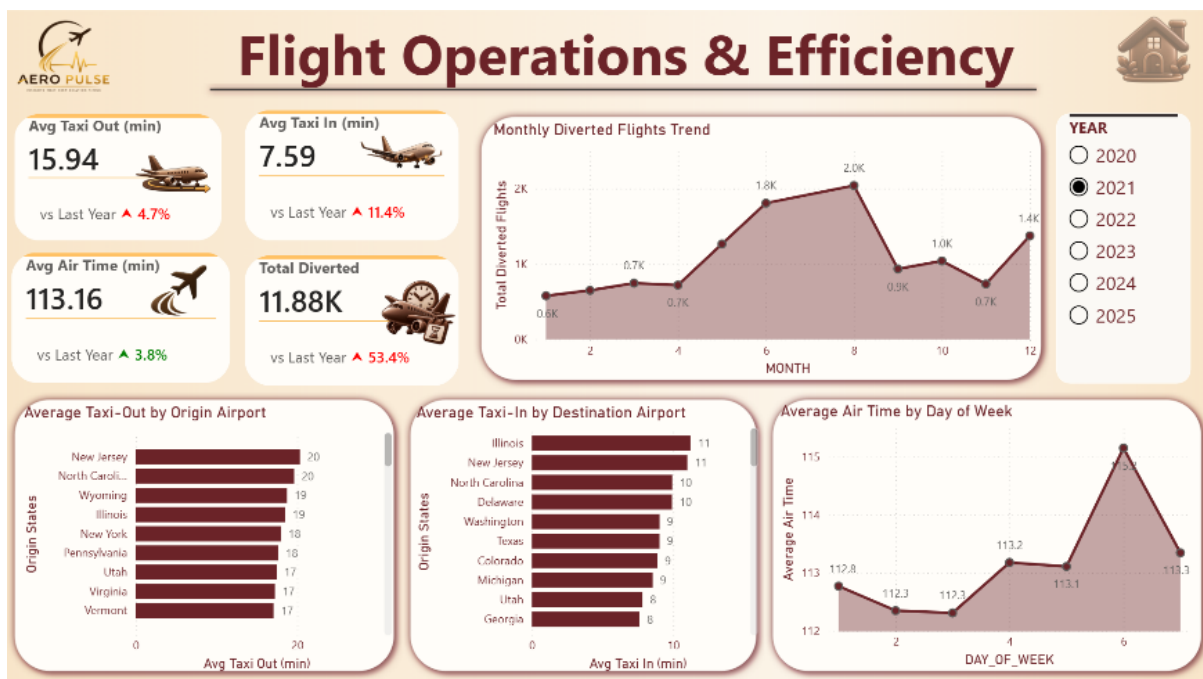
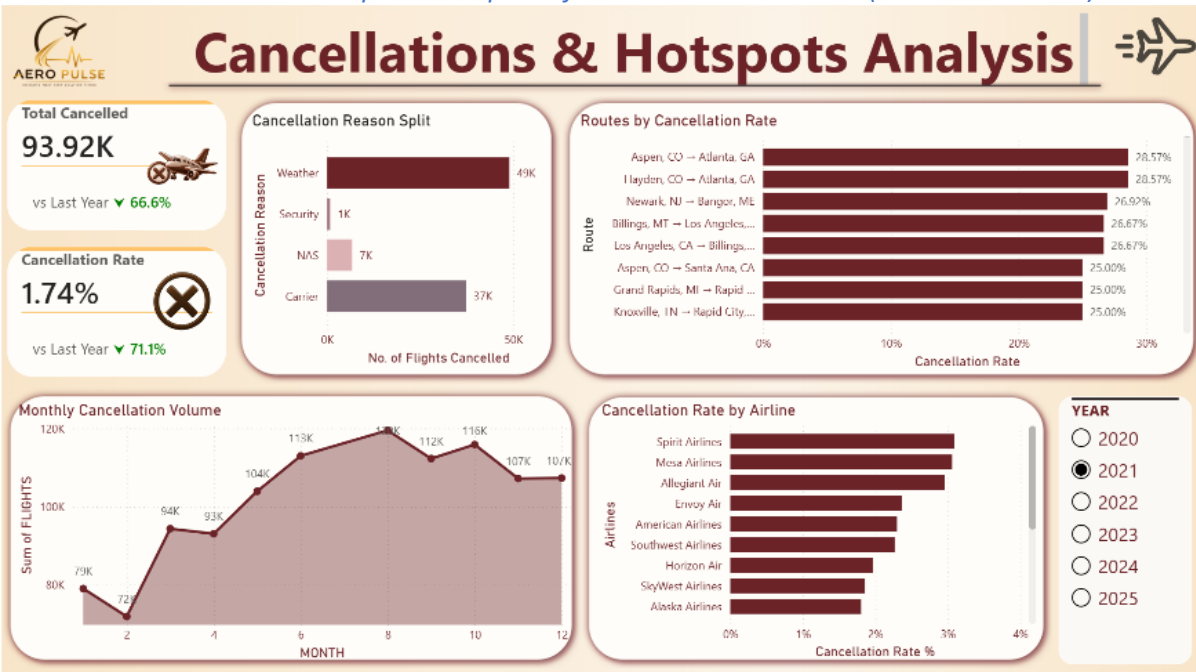


Top 10 Origin Airports by Average Departure Delay



Top 10 Routes by Average Schedule Variance





Explanation:

The Power BI dashboard provides an interactive view of aviation performance and flight delays through KPIs, trends, comparisons, and supporting visualizations. Users can explore the data using the available filters and interact with different visual elements to understand operational patterns.



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Project GitHub Link

The complete project source code, analysis, documentation, and supporting files are available on GitHub.

GitHub Repository:

<https://github.com/sure-trust/HANNE-JENIFER-R-g2-data-analytics>



Learning and Reflection

Learning

Working on AeroPulse provided practical experience across the complete data analytics lifecycle. The key learnings from the project include:

- Improved understanding of **data cleaning, preprocessing, and exploratory data analysis** using Python.
- Gained practical experience in **developing analytical KPIs and interpreting aviation operational data**.
- Strengthened skills in **Power BI dashboard development, data modeling, interactive visualizations, and performance analysis**.
- Gained experience in **website development and presenting analytical results through a user-facing interface**.
- Developed a better understanding of how raw data can be transformed into **meaningful business insights**.
- Improved skills in **project organization, documentation, problem-solving, and communicating technical findings clearly**.

Overall Experience

Developing AeroPulse was a valuable hands-on experience in applying data analytics to a real-world domain. The project provided an opportunity to work through the complete process, from understanding and preparing raw data to analyzing it, building interactive visualizations, and presenting the final insights through a website.

The project also highlighted the importance of data quality, analytical thinking, effective visualization, and clear communication when working with real-world datasets. Overall, AeroPulse strengthened both technical and problem-solving skills while providing practical experience in developing an end-to-end data analytics project.



Conclusion and Future Scope

Conclusion

AeroPulse — Aviation Data Analytics & Operational Insights successfully demonstrates an end-to-end approach to analyzing aviation data and understanding flight delay patterns. The project achieved its objectives by cleaning and analyzing the data, identifying relevant trends and operational insights, developing an interactive Power BI dashboard, and presenting the overall project through a dedicated website.

The project demonstrates how data analytics and business intelligence can transform raw aviation data into clear, interactive, and meaningful insights that support better understanding of operational performance.

Future Scope

AeroPulse can be further enhanced by:

- Incorporating **real-time flight and operational data** for continuous monitoring.
- Developing **predictive models** to forecast potential flight delays.
- Expanding the analysis to include additional operational and external factors such as weather and air traffic conditions.
- Adding more advanced **airline and airport performance benchmarking**.
- Enhancing the website and dashboard with additional interactive analytical features.
- Extending the system to support automated data updates and continuous reporting.